

AMIT KUMAR MAURYA

AI / GEN AI SOFTWARE ENGINEER

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SUMMARY

AI / Software Engineer with 4 years of experience building Python backend systems, **LLM**-powered applications, **AI**-driven automation, APIs, and real-time systems. Experience spans telemetry and crash analysis, **RAG** applications, **FastAPI**/Django services, **PostgreSQL** optimization, secure distributed systems, and low-latency computer-vision pipelines. Targeting GenAI Engineer and Software Engineer roles focused on production AI applications.

TECHNICAL SKILLS

GenAI / LLM: Generative AI, LLMs, RAG, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, LLM APIs, Structured Outputs

AI Frameworks: LangChain, LangGraph, Hugging Face, Transformers, AI Agents, Tool / Function Calling

LLM Engineering: RAG Evaluation, LLM Evaluation, Prompt Evaluation, Hallucination Mitigation, AI Application Design

Backend: Python, FastAPI, Django, REST APIs, SQLAlchemy, Alembic, JWT, RBAC

Data / Database: PostgreSQL, SQL, SQLite

Cloud / DevOps: AWS, Docker, Git, Linux, CI/CD

Real-Time / Vision: OpenCV, GStreamer, MAVLink, PyQt5

TECHNICAL SKILLS

Johnnette Technologies Pvt. Ltd. – Software Engineer | Noida, Uttar Pradesh | Apr 2022 – Present

- Built an AI-assisted drone telemetry analysis workflow for TLOG/DataFlash flight data, using Python and **LLM**-based analysis to turn raw telemetry into structured crash insights and investigate potential factors such as battery, vibration, and flight conditions.
- Developed automated flight-data reporting workflows that transform telemetry into structured diagnostic insights, reducing dependence on manual investigation and supporting faster engineering decisions.
- Built an **LLM**-integrated TLOG analysis pipeline reported to reduce diagnostic time by ~40%, with analysis focused on identifying suspected crash causes and producing human-readable reports.
- Built production **REST** services with **FastAPI** and **Django** for frontend-to-backend communication across distributed systems, including authentication, authorization, and client-facing workflows.
- Implemented **JWT** authentication and RBAC with granular permissions, and developed a centralized license-management service for activation, enforcement, and auditing across distributed installations.
- Optimized **PostgreSQL** schemas and query performance, improving API response times by ~25%.
- Architected a Ground Control System in **PyQt5** and built real-time bidirectional **MAVLink** telemetry for GPS tracking, mission planning, and autonomous flight-path management; operationally deployed with **Indian armed forces** units.
- Integrated **GStreamer** and **OpenCV** into low-latency aerial video workflows for real-time tracking and detection with sub-second latency.

Award: Best UI & Software Development Award — Johnnette Technologies, 2024

SELECTED GENAI & SOFTWARE PROJECTS

P1: WhatRAG — WhatsApp Analysis & RAG Chatbot | *Python · LangChain · RAG · Vector Database · FastAPI*

- Built a RAG-powered chatbot over uploaded WhatsApp chat history, enabling natural-language questions about conversations.
- Implemented analysis for message trends, sentiment, topics, and participant patterns.

P2: FlowForge — GitHub-style Project Management Backend | *FastAPI · PostgreSQL · SQLAlchemy · Alembic · JWT · Docker*

- Built a full CRUD backend with hierarchical boards, lists, and cards, including Alembic migrations and board-level RBAC.

P3: Dev Dashboard — Local Engineering Dashboard | *Python · FastAPI · SQLite · SQLAlchemy · JavaScript · Chart.js*

- Built a unified dashboard for local repositories covering Git activity, project health, tasks, and notes, with automatic sync and a 0–100 health score.

EDUCATION & CERTIFICATION

MCA — Glocal University, 2024 | **BCA** — Glocal University, 2022

Full Stack Generative and Agentic AI with Python — Udemy, 2026

Generative AI Foundations Certificate — UpGrade | Microsoft, 2026